ASSIGNMENT : Advanced Visuals in Tableau

Question 1 : Create a Heatmap to analyze Treatment Cost by Department and Region.

SOLUTION:

Chart Type: Heatmap

Steps:

1. Drag Department to *Rows*
2. Drag Region to *Columns*
3. Drag SUM(TreatmentCost) to *Color*
4. Change Marks type to Square
5. (Optional) Drag SUM(TreatmentCost) to *Label*

Insight:
This heatmap helps identify which department and region combination has the highest treatment cost. Darker colors indicate higher costs.

Question 2 : Create a Highlight Table showing Satisfaction Score by Department.

SOLUTION:

Chart Type: Highlight Table

Steps:

1. Drag Department to *Rows*
2. Drag AVG(SatisfactionScore) to *Color*
3. Use Show Me → Highlight Table

Insight:
This chart highlights departments with high and low satisfaction scores using color intensity.

Question 3 : Create a Bump Chart to show ranking of Departments based on Treatment Cost over months.

SOLUTION :

Chart Type: Bump Chart (Line Chart with Rank)

Steps:

1. Drag MONTH(AdmissionDate) to *Columns*
2. Drag SUM(TreatmentCost) to *Rows*
3. Drag Department to *Color*
4. Right-click SUM(TreatmentCost) → *Quick Table Calculation → Rank*
5. Edit Table Calculation → Compute using Department
6. Change Marks type to Line
7. Reverse axis (Rank 1 at top)

Insight:
Shows how department rankings change over time based on treatment cost.

Question 4 : Create a Butterfly Chart comparing Male vs Female patient count by Department.

SOLUTION :

Chart Type: Butterfly Chart

Steps:

1. Create calculated field:
   Patient Count = COUNT(PatientID)
2. Drag Department to *Rows*
3. Drag Patient Count to *Columns* (twice)
4. For first axis:
   a. Filter Gender = Male
5. For second axis:
   a. Filter Gender = Female
   b. Right-click axis → Reverse
6. Synchronize both axes
7. Apply different colors for Male & Female

Insight:
Compares male vs female patient distribution across departments.

Question 5 : Create a Dual Axis Chart to compare Average Age and Treatment Cost by Department.

SOLUTION :

Chart Type: Dual Axis Chart

Steps:

1. Drag Department to *Columns*
2. Drag AVG(Age) to *Rows*
3. Drag SUM(TreatmentCost) to *Rows*
4. Right-click second axis → Dual Axis
5. Synchronize axes
6. Set Marks:
    a. Age → Line
    b. Cost → Bar

Insight:
Helps compare patient age trends with treatment cost across departments.

Question 6 : Create a Sunburst Chart showing Department → Diagnosis hierarchy by Treatment Cost.

SOLUTION :

Chart Type: Sunburst (using Dual Pie Chart)

Steps:

1. Set Marks type to Pie
2. Drag:
    a. Department → Color
    b. Diagnosis → Detail
    c. SUM(TreatmentCost) → Angle
3. Duplicate the chart
4. Apply Dual Axis
5. Adjust sizes:
    a. Inner circle → Department
    b. Outer circle → Diagnosis

Insight:
Displays hierarchical distribution of treatment cost by department and diagnosis.

Question 7 : Create a Dual Axis Chart comparing Treatment Cost and Satisfaction Score by Department.

SOLUTION :

Chart Type: Dual Axis Chart

Steps:

1. Drag Department to *Columns*
2. Drag SUM(TreatmentCost) to *Rows*
3. Drag AVG(SatisfactionScore) to *Rows*
4. Apply Dual Axis
5. Synchronize axes
6. Set Marks:
    a. Cost → Bar
    b. Satisfaction → Line

Insight:
Shows relationship between cost and patient satisfaction across departments.

Question 8 : Create a Dashboard combining:

Heatmap

Dual Axis Chart

Sunburst Chart

Butterfly Chart

SOLUTION :

Dashboard Components:

- Heatmap
- Dual Axis Chart
- Sunburst Chart
- Butterfly Chart

Steps:

1. Go to **Dashboard → New Dashboard**
2. Drag all created sheets into the dashboard
3. Add filters:
    a. Department
    b. Region
4. Arrange visuals using containers
5. Add dashboard title:
   "Hospital Performance Dashboard"

Insight:
The dashboard provides a comprehensive view of cost, satisfaction, gender distribution, and departmental performance.